Constructing general Dobble games with finite projective planes

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Abstract

Dobble is a card game where each card is marked with eight different symbols and any two cards always share one, and only one, symbol in common. By identifying cards with lines and symbols with points, it turns out that Dobble is a perfect visualization of a geometric structure called finite projective plane of order $p = 7$. For any prime power $q = p^m$, with $m \geq 1$, there exists a finite projective plane of order q, which consists of $q^2 + q + 1$ projective lines and points. Each point is incident with $q + 1$ distinct lines and each line is incident with $q + 1$ distinct points. Any two lines intersect in one and only one point (Dobble property). Can we construct such projective space? In this note we provide, for any prime power $q = p^m$, an explicit formula to arrange $q^2 + q + 1$ distinct symbols into groups of $q+1$ elements that constitute the lines of a projective plane, and therefore the cards of a general Dobble game.

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1 Introduction

Dobble is a game in which players have to find symbols in common between two cards. It uses a deck of 55 circular cards, each printed with 8 different symbols. Any two cards always share one, and only one, matching symbol. This property is referred to as the *Dobble property*. The objective of the game is to be the first player to announce the common symbol between two given cards.



Two Dobble cards, the shared symbol between them is the fire.

In this note we want to answer the general question of how to generate Dobble games with any possible number of symbols and cards. Because the Dobble game is an exact visualization of a finite projective plane, we are interested in generating any possible finite projective plane constructively, which means labeling all its points and grouping them into lines. This study requires finite fields, i.e., fields with a finite number of elements. Finite projective planes are indeed constructed from equivalence classes in vector spaces constructed over finite fields. To make this presentation self-content, these foundational constructions are presented extensively in the first three sections.

Here is the outline in which this work is organized:

Section 2 Introduces the abstract algebra of commutative rings, that includes the fundamental objects and concepts to construct finite fields.

Section 3 Shows how finite fields are explicitly constructed using the algebra of commutative rings and also introduces to finite fields arithmetic.

Section 4 First defines projective planes in general, then shows how finite projective planes are constructed from vector spaces over finite fields.

Section 5 Provides the general procedure to generate finite projective spaces of any prime power order.

Section 6 Shows a Python implementation of the procedure to generate general Dobble games of any prime power order.

The reader already familiar with finite fields and finite projective geometry can directly jump to Section 5.

2 Algebraic fundamentals

Commutative rings generalize the concept of integer operations (addition and multiplication) to more general sets. If the commutative ring R has integrity (0 has no divisors), then prime elements p are those and only those that generate prime ideals (p). The quotient $R/(p)$ is itself integral if and only if (p) is prime. In integral domains primes are also irreducible, but the converse is not generally true. In a Bézout domain the converse is true. Bézout domains are characterized by the property that the greatest common divisor of any two elements can be obtained as linear combination of them. The existence of the Bézout coefficients is not constructive. Euclidian domains are integral domains in which the Euclidian division works in the same way as for integers. Euclidian domains are in particular Bézout domains, but the Bézout coefficients can be explicitly obtained by iterative application of the Euclidian division algorithm. In an Euclidian domain the quotient with an ideal generated by an irreducible element is an integral domain, if the quotient is also finite, then it is a finite field.

2.1 Commutative rings and ideals

Definition 2.1 (Commutative ring)**.** *A set R equipped with two operations $+, \cdot : R \times R \to R$ forms a **commutative ring** when these operations reflect certain familiar properties of real, rational or integer numbers. It must be:*

R commutative group under addition $(+)$:

- ***Associativity:*** $(a + b) + c = a + (b + c)$
- ***Identity:*** $\exists z : z + a = a + z = a,\ \forall a$ *(z usually denoted 0)*
- ***Inverse:*** $\forall a\ \exists y : a + y = y + a = z$ *(y usually denoted $-a$)*
- ***Commutativity:*** $a + b = b + a,\ \forall a, b$

R monoid under multiplication $(\cdot)$:

- ***Associativity:*** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
- ***Identity:*** $\exists u : u \cdot a = a \cdot u = a,\ \forall a$ *(u usually denoted 1)*
- ***Commutativity:*** $a \cdot b = b \cdot a,\ \forall a, b$

R distributive of multiplication over summation:

- ***Distributivity:*** $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

Definition 2.2 (Ideal)**.** *A subset $I \subseteq R$ (proper subset $I \subset R$) of a commutative ring R is called **ideal** (**proper ideal**) if:*

I is closed under addition $(+)$ and absorbs multiplication $(\cdot)$:

- ***Additive subgroup:*** $a, b \in I \implies a - b \in I$
- ***Absorption:*** $a \in I, b \in R \implies a \cdot b \in I$

Definition 2.3 (Ideal sum)**.** *Let $I, J \subseteq R$ be ideals of a commutative ring R. The **ideal sum** of I and J is the set $I + J := \{i + j : i \in I, j \in J\}$.*

Corollary 2.1. *The ideal sum $I + J$ is an ideal.*

Definition 2.4 (Principal ideal)**.** *An ideal $I \subseteq R$ of a commutative ring R is called a **principal ideal** if there exists $b \in R$ such that $I =: (b) := \{b \cdot x : x \in R\} = \{x \cdot b : x \in R\}$.*

Definition 2.5 (Prime ideal)**.** *A proper ideal $P \subset R$ of a commutative ring R is called a **prime ideal** if:*

$R \setminus P$ is closed under multiplication $(\cdot)$:

- ***Primality:*** $a, b \notin P \implies a \cdot b \notin P$

Definition 2.6 (Integral domain)**.** *A commutative ring D is an **integral domain** if:*

$D \setminus \{0\}$ is closed under multiplication $(\cdot)$:

- ***Integrity:*** $a, b \neq 0 \implies a \cdot b \neq 0$

Definition 2.7 (Principal ideal domain)**.** *An integral domain is a **principal ideal domain** if every ideal is a principal ideal.*

Definition 2.8 (Prime element)**.** *Let R be a commutative ring. A non-zero non-unit element $p \in R$ is a **prime element** if for every $a, b \in R$ such that $p|ab$, then $p|a$ or $p|b$.*

Lemma 2.1. *Let R be a commutative ring. An element $p \in R$ is prime if and only if the ideal (p) is prime.*

Proof. ($\implies$) Let $p \in R$ prime. Let $a, b \in pR$, then clearly $p|ab$. By assumption of p prime we have either $p|a$ or $p|b$, thus we would have either $a \in (p)$ or $b \in (p)$.

($\impliedby$) Let (p) be prime ideal. Let $a, b \in D$ such that $p|ab$, we then have $ab \in (p)$ and by assumption of prime ideal it is either $a \in (p)$ or $b \in (p)$. We would either have then $p|a$ or $p|b$. □

2.2 Bézout and Euclidian domains

Definition 2.9 (Bézout domain)**.** *An integral domain D is a **Bézout domain** if the sum of any two principal ideals is also a principal ideal.*

Corollary 2.2. *A principal ideal domain is a Bézout domain.*

Definition 2.10 (Greatest common divisor)**.** *Let R be a commutative ring. The **greatest common** divisor between two elements $a, b \in R$ is an element $d = gcd(a, b) \in R$ such that:*

- *$d|a$*
- *$d|b$*
- *if $c \in R$ is such that $c|a$ and $c|b$, then also $c|d$.*

Theorem 2.1. *An integral domain D is a Bézout domain if and only if for every $a, b \in D$ there exist $x, y \in D$ such that $d = gcd(a, b) = ax + by$.*

Proof. ($\Longrightarrow$) Let D be Bézout domain. Given any two $a, b \in D$, the two ideals (a) and (b) are principal. By assumption their ideal sum is also principal, thus there exists $d \in D$ such that $(d) = (a) + (b)$. This means there exist $x, y \in D$ such that $d = ax + by$. Let's show that $d = gcd(a, b)$.

- $a \in (a) \subseteq (a) + (b) = (d)$, then $d|a$.
- $b \in (b) \subseteq (a) + (b) = (d)$, then $d|b$.
- Let $c \in D$ such that $c|a$ and $c|b$, then $c|ax + by = d$.

($\Longleftarrow$) Let D be integral domain and $I, J \subseteq D$ two principal ideals. Then there exist $a, b \in D$ such that $I = (a)$ and $J = (b)$. By assumption there exist $x, y \in D$ such that $d = gcd(a, b) = ax + by$. Let us prove that $(d) = (a) + (b)$.

($\subseteq$) Let $z \in (d)$, then there exists $w \in D$ such that $z = dw = (ax + by)w = axw + byw \in (a) + (b)$.

($\supseteq$) Let $z \in (a) + (b)$, then there exist $h, k \in D$ such that $z = ah + bk$. By definition of greatest common divisor we have that $d|a$ and $d|b$ and so there exist $s, t \in D$ such that $a = ds$ and $b = dt$. We have $z = ah + bk = dsh + dtk = d(sh + tk) \in (d)$.

□

Definition 2.11 (Irreducible element)**.** *Let R be a commutative ring. A non-zero non-unit element $p \in R$ is an **irreducible element** if for every $a, b \in R$ such that $p = ab$, then a or b is invertible (being the other an associate of p).*

Lemma 2.2. *Let D be an integral domain. Any $p \in D$ prime is also irreducible.*

Proof. Let $a, b \in D$ such that $p = ab$. Trivially $ab = p \cdot 1$, thus $p|ab$. By primality it is either $p|a$ or $p|b$. By symmetry we can assume there exists $s \in D$ such that $a = ps$, thus $p = ab = psb$ and so $p(1 - sb) = 0$. By integrity $1 - sb = 0$ and so b is invertible. □

Lemma 2.3. *Let D be a Bézout domain. Then $p \in D$ is prime if and only if is irreducible.*

Proof. Let $p \in D$ be irreducible and let $a, b \in D$ such that $p|ab$, thus $ab = pk$ for some $k \in D$. If $p|a$ we are done. Let then be $p \nmid a$. By Bézout property there exists $d \in D$ such that $(p) + (a) = (d)$. Given $p \in (p) \subset (p) + (a) = (d)$, we have $d|p$. Because of irreducibility, it must be d associate of p, or invertible. In the first case we would have $(p) = (d) = (p) + (a)$ and so $p|a$, which is not possible. Then d is invertible and thus $(p) + (a) = (d) = R$ and there exist $x, y \in D$ such that $1 = px + ay$. Multiplying by b we get $b = bpx + aby = p(bx + ky)$, thus $p|b$. □

Corollary 2.3. *Let D be a Bézout domain. Then $p \in D$ is irreducible if and only if (p) is prime ideal.*

Definition 2.12 (Euclidian function)**.** *Let D be an integral domain. A function $N : D \to \mathbb{N}\setminus\{0\}$ is an **Euclidian function** if it satisfies the following property: for any $a, b \in D$ with $b \neq 0$, there exist $q, r \in D$ such that $a = bq + r$ with either $r = 0$ or $N(r) < N(b)$.*

Definition 2.13 (Euclidian domain)**.** *An integral domain is an **Euclidian domain** if it admits at least one Euclidian function.*

Theorem 2.2. *An Euclidian domain is a principal ideal domain.*

Proof. Let $I \subset E$ be an ideal of an Euclidian domain E. If $I = \{0\}$ we are done. Assume $I \neq \{0\}$. Let $N : E \to \mathbb{N}$ be an Euclidian function and let $S := \{N(i) : i \in I \setminus \{0\}\}$ be the image of this map over the non zero elements of I. The set S is a non-empty subset of the non-negative integers, by the well-ordering principle there exists $b \in I$ such that $N(b) \leq N(i)$ for every $i \in I \setminus \{0\}$. Let us prove that $(b) = I$:

($\subseteq$) By assumption of ideal $b \cdot x \in I$ for every $x \in E$.

($\supseteq$) Let $y \in I$. There exist $q, r \in E$ such that $y = bq + r$ with either $r = 0$ or $N(r) < N(b)$. Then we have $r = y - bq \in I$ by the properties of ideals. If we assume $r \neq 0$, then $N(r) \in S$, which means $N(r) \geq N(b)$ by minimality of $N(b)$ in S. This is a contradiction. It must be $r = 0$, and therefore $y = bq \in (b)$. □

2.3 Quotient rings and fields

Definition 2.14 (Field)**.** *A **field** F is a commutative ring where all its non zero elements are invertible under multiplication, therefore:*

$F \setminus \{0\}$ commutative group under multiplication $(\cdot)$:

- ***Inverse:*** $\forall a\ \exists w : a \cdot w = w \cdot a = u$ *(w usually denoted a^{-1})*

Corollary 2.4. *A finite integral domain is a field.*

Proof. Let D be an integral domain and $a \in D$ with $a \neq 0$. We need to prove that a^{-1} exists. Let's consider $\lambda_a : D \to D : x \mapsto a \cdot x$ the left multiplication by a as a map. This map is injective because for any $x, y \in D$ such that $\lambda_a(x) = \lambda_a(y)$ we have $0 = a \cdot x - a \cdot y = a \cdot (x - y)$. By integrity it can only be $x - y = 0$, thus $x = y$. By finiteness of D, the map λ_a is also surjective. The claim is proved because $1 \in D$ admits existence of $b \in D$ such that $1 = \lambda_a(b) = a \cdot b$, it's $b = a^{-1}$. □

Definition 2.15 (Quotient set)**.** *Let R be a commutative ring and $I \subset R$ a proper ideal. Let $\sim_I$ be an equivalence relation on R such that for any $x, y \in R$ it is $x \sim_I y$ if and only if $x - y \in I$. The equivalence class of an element $x \in R$ with respect to $\sim_I$ is defined as $[x]_I := \{y \in R : x - y \in I\}$. Notice that $x \sim_I y$ if and only if $[x]_I = [y]_I$, and also $I = [0]_I$. The **quotient set** of R by I is notated R/I and is defined as the set of all equivalence classes w.r.t. $\sim_I$ on R, formally $R/I := \{[x]_I : x \in R\}$.*

Corollary 2.5 (Quotient ring)**.** *The quotient set R/I is a commutative ring under the multiplication and addition defined by:*

- $[x]_I \cdot [y]_I = [x \cdot y]_I$

- $[x]_I + [y]_I = [x + y]_I$

Notice that the identity for addition is $[0]_I$ *and for multiplication is* $[1]_I$.

Proof. It is an easy exercise to verify that these operations are well defined and support the structure of a ring. □

Theorem 2.3 (Quotient ring test)**.** *Let* R *be a commutative ring. A proper ideal* $P \subset R$ *is prime if and only if the quotient ring* R/P *is an integral domain.*

Proof. ($\implies$) Assume $P \subset R$ is prime ideal. We only need to prove that R/P has no divisors for $[0]_P$. Let $[x]_P, [y]_P \neq [0]_P$ in R/P. We clearly have $x, y \notin P$, and since P is prime ideal it is also $x \cdot y \notin P$. In conclusion we have $[x]_P \cdot [y]_P = [x \cdot y]_P \neq [0]_P$ and the claim is proved.

($\impliedby$) Assume R/P is integral domain. We need to prove that $R \setminus P$ is closed under multiplication. Let $x, y \notin P$, then clearly $[x]_P, [y]_P \neq [0]_P$. Being $R \setminus P$ integral domain we have $[0]_P \neq [x]_P \cdot [y]_P = [x \cdot y]_P$, therefore $x, y \notin P$ and we are done. □

Lemma 2.4. *Let* E *be an Euclidian domain and* $p \in E$ *irreducible. Then* $E_p := E/(p)$ *is an integral domain and for every* $a \in E$ *we have* $[a]_{E_p} = [r]_{E_p}$*, where* $r \in E$ *is the remainder of* a *after division by* p*.*

Proof. By Euclid's property we have $a = pb + r$ with either $r = 0$ or $N(r) < N(b)$. Let us prove the set equivalence by both inclusions.

($\subset$) Let $x \in [a]_{E_p}$, then there exists $z \in E$ such that $a - x = pz$. Adding together we have $r - x = (a - pb) + (pz - a) = pz - pb = p(z - b)$, thus $x \in [r]_{E_p}$.

($\supset$) Let $y \in [r]_{E_p}$, then there exists $z \in E$ such that $r - y = pz$. Adding together we have $a - y = (pb + r) + (pz - r) = pb + pz = p(b + z)$, thus $y \in [a]_{E_p}$. □

3 Finite fields

Finite fields are the main subject of a vast branch of mathematics known as Galois theory, named after the French mathematician Evariste Galois, who laid its foundations before dying at the young age of 20. It is a well-known result of Galois theory that finite fields exist and the order of a finite field can only be a power p^m of a prime number p, with $m \geq 1$. Conversely, for any prime power $q = p^m$, with $m \geq 1$, there always exists a finite field with q elements. All finite fields of the same order q are isomorphic, therefore justifying the uniform notation $\mathbb{F}_q$ for a field of order q. A finite field $\mathbb{F}_p$ whose order is a prime number p can be constructed canonically from the commutative ring $\mathbb{Z}$ of integer numbers by quotient with respect to the ideal of multiples of p. Equivalence classes are indeed the reminders after division by p. More generally, a finite field $\mathbb{F}_q$ whose order is a prime power $q = p^m$ can be constructed canonically from the commutative ring $\mathbb{Z}_p[X]$ of polynomials with coefficients in the finite field $\mathbb{Z}_p$ by quotient with respect to the ideal of multiples of an irreducible polynomial $w(X)$ of degree m. Equivalence classes are indeed the reminders after polynomial division by $w(X)$. These *canonical* constructions are revised in the following sections.

Definition 3.1. *For any positive integer* $n > 1$ *we define* $\overline{n} := \{0, \ldots, n - 1\}$.

3.1 Fields of prime order

Let us consider $\mathbb{Z}$ the set of integer numbers. We give as known fact that $\mathbb{Z}$ is a commutative ring with the usual operations of addition and multiplication. Since we are also familiar from elementary school that $a \cdot b \neq 0$ implies both $a, b \neq 0$, we also take for granted that $\mathbb{Z}$ is an integral domain.

Theorem 3.1 (Euclidian domain $\mathbb{Z}$)**.** *For any $a, b \in Z$ with $b \neq 0$, there exist $q, r \in \mathbb{Z}$ such that $a = qb + r$ with $0 \leq r < |b|$.*

Let now be $n \in \mathbb{Z}$ a positive integer $n > 1$. The ideal $(n) := \{nz : z \in \mathbb{Z}\}$ is a principal ideal in $\mathbb{Z}$. Because $\mathbb{Z}$ is an Euclidian domain it is in particular a Bézout domain, by Corollary 2.3 we then have that $p \in \mathbb{Z}$ irreducible (i.e., prime as defined for integers) implies (p) is prime ideal in $\mathbb{Z}$. Because of the Quotient ring test (Theorem 2.3) the quotient ring $\mathbb{Z}_p := \mathbb{Z}/(p)$ is an integral domain. By Lemma 2.4 equivalence classes in $\mathbb{Z}_p$ can be identified with the set of the remainders after Euclidian division by p. This justifies the classical identification $\mathbb{Z}_p = \overline{p} = \{0, 1, \ldots, p-1\}$. By finiteness, $\mathbb{Z}_p$ is a field.

Corollary 3.1. *$\mathbb{F}_p := \mathbb{Z}_p$ is a finite field (of p elements).*

Proof. $\mathbb{Z}_p$ is a finite integral domain, thus a field by Corollary 2.4. □

In $\mathbb{F}_p$ the arithmetic is p-modular. For every $a, b \in \mathbb{Z}$, let $r_a, r_b \in \{0, \ldots, p-1\}$ be their remainders after division by p, then:

$$[a] + [b] = [r_a] + [r_b] = [r_a + r_b]$$
$$[a][b] = [r_a][r_b] = [r_a r_b]$$

This shows that arithmetic is unambiguously performed in $\mathbb{F}_p$ on the remainder values. For the sake of illustration, consider arithmetic tables for $\mathbb{F}_5$:

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\cdot$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

3.2 Fields extensions

Let p be prime, we denote by $\mathbb{Z}_p[X]$ the commutative ring of polynomials with coefficients in the finite field $\mathbb{Z}_p$. A polynomial $f(X) \in \mathbb{Z}_p[X]$ is an expression of the form $f(X) = \sum_{i=0}^{n} f_i X^i$ with $f_i \in \mathbb{Z}_p$. The degree of a polynomial $f(X)$ is the largest power in which the symbol X appears in the expression with corresponding non-zero coefficient, compactly $\deg(f) = \max\{i : f_i \neq 0\}$. Summation, difference, multiplication, and division are performed in $\mathbb{Z}_p[X]$ by p-modular arithmetic on the polynomial coefficients.

Theorem 3.2 (Euclidian domain $\mathbb{Z}_p(X)$)**.** *Let $a(X), b(X) \in \mathbb{Z}_p[X]$ with $b(X) \not\equiv 0$, there exist $g(X), r(X) \in \mathbb{Z}_p[X]$ such that $f(X) = g(X)b(X)+r(X)$ with either $r \equiv 0$ or $\deg(r) < \deg(g)$.*

Let now be $w(X) \in \mathbb{Z}_p[X]$ a polynomial with $\deg(w) > 1$. The ideal $(w(X))$ is a principal ideal in $\mathbb{Z}_p[X]$. Because $\mathbb{Z}_p[X]$ is an Euclidian domain it is in particular a Bézout domain, by Corollary 2.3 we then have that $w(X)$ irreducible implies $(w(X))$ is prime ideal. Let us consider any integer $m > 1$, it is a result of Galois theory that there always exists an irreducible polynomial $w(X) \in \mathbb{Z}_p[X]$ with $\deg(z) = m$. Because of the Quotient ring test (Theorem 2.3) the quotient ring $\mathbb{Z}_p[X]/(w(X))$ is an integral domain. By Lemma 2.4 equivalence classes in $\mathbb{Z}_p[X]/(w(X))$ can be identified with the set of the remainders after Euclidian division by $w(X)$. These are clearly all and only the polynomials whose degree is strictly less than m. Because for any degree $k = 0, \ldots, m-1$ there are p distinct choices of coefficients, this set consists of exactly $q := p^m$ distinct elements.

Corollary 3.2. *$\mathbb{F}_q := \mathbb{Z}_p[X]/(w(X))$ is a finite field (of $q = p^m$ elements).*

Proof. $\mathbb{F}_p$ is a finite integral domain, thus a field by Corollary 2.4. □

In $\mathbb{F}_q$ the arithmetic is $w(X)$-modular. For every $a(X), b(X) \in \mathbb{Z}_p[X]$, let $r_a(X), r_b(X) \in \mathbb{Z}_p[X]$ be their remainders after division by $w(X)$, then:

$$[a(X)] + [b(X)] = [r_a(X)] + [r_b(X)] = [r_a(X) + r_b(X)]$$
$$[a(X)][b(X)] = [r_a(X)][r_b(X)] = [r_a(X) r_b(X)]$$

This shows that arithmetic is unambiguously performed in $\mathbb{F}_q$ on the remainder values. However, to visualize them as tabular operations on the integers $\overline{q} = \{0, \ldots, q-1\}$ a bijection is needed.

Theorem 3.3. *The elements of the field $\mathbb{F}_q$ can be put in bijective correspondence with the set $\overline{q} = \{0, \ldots, q-1\}$ via the radix-p evaluation map:*

$$\varphi \colon \mathbb{F}_q \longrightarrow \overline{q} : [r(X)] \longmapsto r(p)$$

with inverse

$$\psi \colon \overline{q} \longrightarrow \mathbb{F}_q : n \longmapsto \left[\sum\nolimits_{k=0}^{m-1} r_k X^k \right]$$

where the coefficients $r_k \in \overline{p}$ are the remainders generated by the iterative Euclidian division by p, starting from n.

Proof. This is true because any integer $0 \leq n < q$ has unique radix-p representation of the form $n = \sum_{k=0}^{m-1} r_k p^k$, where the coefficients $r_k \in \overline{p} = \{0, \ldots, p-1\}$ are obtained as remainders from the iterative division process:

$$\begin{aligned} n = & a_0 = a_1 p + r_0 \\ & a_1 = a_2 p + r_1 \\ & \ldots \\ & a_{m-1} = a_m p + r_{m-1} \\ & a_m = 0 \end{aligned}$$

The integer can be reconstructed as: $n = a_0 = a_1 p + r_0 = (a_2 p + r_1)p + r_0 = a_2 p^2 + r_1 p + r_0 = \ldots = 0 \cdot p^m + r_{m-1} p^{m-1} + \ldots + r_2 p^2 + r_1 p + r_0$. □

Notice that, up to a permutation of the integers, the arithmetic does not change if we replace the irreducible polynomial with another one of the same degree. For the sake of illustration, we show here how to perform arithmetic in $\mathbb{F}_8$. We have $8 = 2^3$, thus $p = 2$ and $m = 3$. The polynomial $w(X) = X^3 + X + 1$ is irreducible of $deg(w) = 3$ in $\mathbb{Z}_2[X]$. In order to get the result of an integer operation between $a, b \in \overline{8} = \{0, 1, 2, 3, 4, 5, 6, 7\}$, we have to perform the operation among the two polynomials $\psi(a)$ and $\psi(b)$, compute $r(X)$ the remainder of the result after polynomial division by $w(X)$, and map it back to an integer by $\varphi(r(X))$. First of all, we clearly have:

a	$\psi(a)$	a	$\psi(a)$
0	0	4	X^2
1	1	5	$X^2 + 1$
2	X	6	$X^2 + X$
3	$X + 1$	7	$X^2 + X + 1$

The remainder after division by $w(X)$ for polynomial operations among these are:

+	0	1	X	$X+1$	X^2	X^2+1	X^2+X	X^2+X+1
0	0	1	X	$X+1$	X^2	X^2+1	X^2+X	X^2+X+1
1	1	0	$X+1$	X	X^2+1	X^2	X^2+X+1	X^2+X
X	X	$X+1$	0	1	X^2+X	X^2+X+1	X^2	X^2+1
$X+1$	$X+1$	X	1	0	X^2+X+1	X^2+X	X^2+1	X^2
X^2	X^2	X^2+1	X^2+X	X^2+X+1	0	1	X	$X+1$
X^2+1	X^2+1	X^2	X^2+X+1	X^2+X	1	0	$X+1$	X
X^2+X	X^2+X	X^2+X+1	X^2	X^2+1	X	$X+1$	0	1
X^2+X+1	X^2+X+1	X^2+X	X^2+1	X^2	$X+1$	X	1	0

·	0	1	X	$X+1$	X^2	X^2+1	X^2+X	X^2+X+1
0	0	0	0	0	0	0	0	0
1	0	1	X	$X+1$	X^2	X^2+1	X^2+X	X^2+X+1
X	0	X	X^2	X^2+X	$X+1$	1	X^2+X+1	X^2+1
$X+1$	0	$X+1$	X^2+X	X^2+1	X^2+X+1	X^2	1	X
X^2	0	X^2	$X+1$	X^2+X+1	X^2+X	X	X^2+1	1
X^2+1	0	X^2+1	1	X^2	X	X^2+X+1	$X+1$	X^2+X
X^2+X	0	X^2+X	X^2+X+1	1	X^2+1	$X+1$	X	X^2
X^2+X+1	0	X^2+X+1	X^2+1	X	1	X^2+X	X^2	$X+1$

Back to integer representatives, we have arithmetic tables for $\mathbb{F}_8$:

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	0	3	2	5	4	7	6
2	2	3	0	1	6	7	4	5
3	3	2	1	0	7	6	5	4
4	4	5	6	7	0	1	2	3
5	5	4	7	6	1	0	3	2
6	6	7	4	5	2	3	0	1
7	7	6	5	4	3	2	1	0

·	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	3	1	7	5
3	0	3	6	5	7	4	1	2
4	0	4	3	7	6	2	5	1
5	0	5	1	4	2	7	3	6
6	0	6	7	1	5	3	2	4
7	0	7	5	2	1	6	4	3

4 Finite projective geometry

A geometric structure consists of a set of points and lines, and a relation between points and lines called *incidence*. A geometric structure is called *projective plane* when the incidence relation satisfies the following properties:

Axiom 1: [No ellipticity] For any two distinct points there is exactly one, and only one, line incident with both of them.

Axiom 2: [No parallels] For any two distinct lines there is exactly one, and only one, point incident with both of them.

Axiom 3: [No degeneracy] There exists four points with no line incident with more than two of them.

The first axiom implies no spherical geometries: infinitely many lines are incident with two points on a sphere, just like all meridians cross both poles of the Earth. The second axiom implies there are no parallel lines on a projective plane. This is particularly important because, after identifying cards with lines and symbols with points, it turns out this axiom perfectly resembles the Dobble property. The third axiom excludes degenerate cases like single points or lines, a single point incident with all the lines, etc. The incidence relation is symmetric with respect to its terms, therefore the expression "point P is incident with line ℓ" replaces both "point P lies on line ℓ" and "line ℓ passes through point P".

4.1 Projective planes

The common approach to construct a projective plane is by endorsing a field vector space of dimension three $V = F^3$ with an equivalence relation for its linear subspaces (vector lines and planes through the origin). Then, points of the projective plane are in correspondence with vector lines, and, projective lines are in correspondence with vector planes.

Definition 4.1 (Projective plane $\mathbb{P}^2$ over the field F)**.** *Let's define the two equivalence relations for any $x, y \in F^3$:*

- *Vector lines:* $x \sim \lambda x \quad \forall \lambda \in F \setminus \{0\}$.
- *Vector planes:* $(x, y) \sim (\alpha x + \beta y, \gamma x + \delta y) \quad \forall (\alpha, \beta), (\gamma, \delta) \in F^2 \setminus \{(0, 0)\}$.

Projective points and lines are defined as equivalence classes on $F^3 \setminus \{(0, 0, 0)\}$ modulo the two equivalence relations for vector lines and planes, respectively. Thus $p = [x]$ for all $x \in F^3$, and $\ell = [(x, y)]$ for all $x, y \in F^3$.

Corollary 4.1. *The projective plane $\mathbb{P}^2$ verifies the axioms.*

Proof.

1. Let $p, q \in \mathbb{P}^2$ be distinct points. Then $p = [x]$ and $q = [y]$ with $x, y \in F^3$ not proportional. It is then well defined and unique the vector plane that includes both of them $\langle x, y \rangle \subset F^3$. The projective line $\ell = [(x, y)]$ is therefore the unique line such that $p, q \in \ell$.

2. Let $\ell_1, \ell_2 \in \mathbb{P}^2$ be distinct lines. Then $\ell_1 = [(x, y)]$ and $\ell_2 = [(u, v)]$ with either the first or the second coordinates not proportional, or both. It is then well defined and unique the vector line which is the intersection $\langle x, y \rangle \cap \langle u, v \rangle$. Let $z \in F^3 \setminus \{(0, 0, 0)\}$ such that $z \in \langle x, y \rangle \cap \langle u, v \rangle$. The projective point $p = [z]$ is therefore the unique point such that $p \in \ell_1 \cap \ell_2$.

3. The axiom is verified by finding four vector lines such that no vector plane includes more than two of them. Let us consider the four points $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ and $e_4 = (1, 1, 1)$ in F^3. If

a vector plane includes a point then it also includes the vector line of its proportional points. Assume there exists a vector plane that includes three of these points (and thus their generated vector lines), then the three points would be linearly dependent. But no group of three of these points are linearly dependent.

□

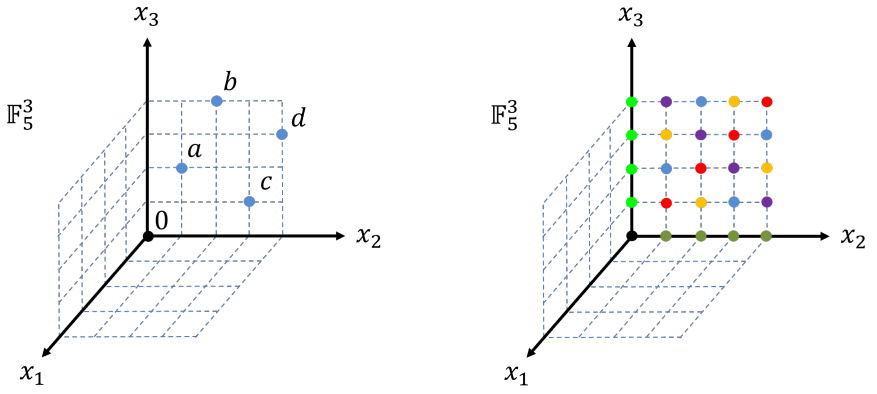
4.2 Finite projective planes

Let p be prime and $m \geq 1$ and consider the finite field $\mathbb{F}_q$ with $q = p^m$. We construct a 3-dimensional vector space from the cartesian product $\mathbb{F}_q^3 := \mathbb{F}_q \times \mathbb{F}_q \times \mathbb{F}_q$ by the following vector addition and scalar multiplication operations:

- $x + y = (x_1, x_2, x_3) + (y_1, y_2, y_3) := (x_1 + y_1, x_2 + y_2, x_3 + y_3)$
- $ax = a(x_1, x_2, x_3) := (ax_1, ax_2, ax_3)$

for every $x, y \in \mathbb{F}_q^3$ and $a \in \mathbb{F}_q$. These operations are clearly in $\mathbb{F}_q$ arithmetic.

Let $\mathbb{P}_q^2$ be the finite projective plane whose points and lines are obtained as equivalence classes on lines and planes in $\mathbb{F}_q^3$. We can visualize with a simple example how vector lines and planes look like in $\mathbb{F}_q^3$. Let us fix $q = 5$, and the point $\boldsymbol{a} = (0, 1, 2)$ laying in the vector plane $(x_1 = 0) = \{x \in \mathbb{F}_5^3 : x_1 = 0\}$. The vector line generated by $\boldsymbol{a}$ is $\langle \boldsymbol{a} \rangle = \{\lambda \boldsymbol{a} : \lambda \in \mathbb{F}_5\}$, it consists of the points: $\boldsymbol{0} = 0\boldsymbol{a} = (0, 0, 0)$, $\boldsymbol{a} = 1\boldsymbol{a} = (0, 1, 2)$, $\boldsymbol{b} = 2\boldsymbol{a} = (0, 2, 4)$, $\boldsymbol{c} = 3\boldsymbol{a} = (0, 3, 1)$, and $\boldsymbol{d} = 4\boldsymbol{a} = (0, 4, 3)$, see figure below, on the left. The other lines in the plane $(x_1 = 0)$ can be generated in a similar manner, see figure below, on the right. Notice that there are $\frac{q^2-1}{q-1} = 6$ lines on the vector plane and $q - 1 = 4$ non-zero points on each line. This is true in general for every line and every plane in $\mathbb{F}_q^3$.



Vector lines in $\mathbb{F}_5^3$. On the left, the line generated by point $\boldsymbol{a} = (0, 1, 2)$. On the right all lines included in the vector plane $(x_1 = 0)$, marked with different colors. The origin is common to all vector lines.

4.3 Counting points and lines

Lemma 4.1. *In $\mathbb{F}_q^3 \setminus \{\boldsymbol{0}\}$ there are exactly $q - 1$ distinct points on every vector line and $q^2 - 1$ distinct points on every vector plane.*

Proof. A vector line is the linear space $L = \langle \boldsymbol{x} \rangle$ generated by a non-zero point $\boldsymbol{x} \in \mathbb{F}_q^3$, thus $L = \{\lambda \boldsymbol{x} : \lambda \in \mathbb{F}_q\}$. Every generated element $\lambda \boldsymbol{x}$ is distinct from the others. Let's suppose that there are $\lambda_1, \lambda_2 \in \mathbb{F}_q$ both non-zero, with $\lambda_1 \neq \lambda_2$ and such that $\lambda_1 \boldsymbol{x} = \lambda_2 \boldsymbol{x}$, then we have $(\lambda_1 - \lambda_2)\boldsymbol{x} = \boldsymbol{0}$ and so $\lambda_1 - \lambda_2 = 0$, which is absurd. Therefore, there are exactly $q-1$ distinct non-zero points in L. A similar argument works for a vector plane, which is the linear space generated by two independent (i.e., non proportional) non-zero points $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{F}_q^3$, that is $P = \{\lambda \boldsymbol{x} + \mu \boldsymbol{y} : \lambda, \mu \in \mathbb{F}_q\}$. There are indeed $q^2 - 1$ distinct non-zero (λ, μ) pairs generating P. □

Taking advantage of the previous result and the correspondence between vector lines and planes and projective points and lines, we can count how many points there are in the projective points and lines, and in the whole projective plane as well.

Lemma 4.2. *In $\mathbb{P}_q^2$ there are exactly $q+1$ distinct points on every projective line and each point is on $q+1$ distinct projective lines.*

Proof. To count the number of distinct points there are on a projective line in $\mathbb{P}_q^2$ we can equivalently count the number of distinct vector lines there are on a vector plane in $\mathbb{F}_q^3$. The point at the origin belongs to all vector lines and planes. Every two vector lines intersect only at the origin. Therefore, we can count inside linear planes and lines by excluding the origin. By Lemma 4.1, a vector plane contains $q^2 - 1$ points and a vector line contains $q - 1$ points. Therefore, in a vector plane there are $\frac{q^2-1}{q-1} = q + 1$ lines. To count the number of distinct projective lines there are on a point in $\mathbb{P}_q^2$ we can equivalently count the number of distinct vector planes there are on a vector line in $\mathbb{F}_q^3$. Again, we can count inside vector spaces by excluding the origin. Let us fix a vector line, i.e., $q - 1$ points. All the remaining points in the whole space are thus $(q^3 - 1) - (q - 1)$. Among them, every vector plane contains $(q^2 - 1) - (q - 1)$ points. Therefore there are $\frac{(q^3-1)-(q-1)}{(q^2-1)-(q-1)} = q + 1$ distinct planes on the fixed line. □

Lemma 4.3. *In $\mathbb{P}_q^2$ there are exactly $q^2 + q + 1$ distinct points and lines.*

Proof. To count the number of distinct points and lines there are in $\mathbb{P}_q^2$ we can equivalently count the number of distinct vector lines and planes there are in $\mathbb{F}_q^3$. Just like before, we can count inside linear planes and lines by excluding the origin. In the whole linear space there are $q^3 - 1$ points, with $q - 1$ points belonging to each line, thus there are $\frac{q^3-1}{q-1} = q^2 + q + 1$ distinct lines in $\mathbb{F}_q^3$. All choices of two points, the first one picked among $q^3 - 1$ and the second one among all $(q^3 - 1) - (q - 1) = q^3 - q$ non-proportional ones, correspond to vector planes. To count the distinct ones, we need to divide by the number of such choices that correspond to the same plane. In each plane we count them as all choices of two points, the first one picked among $q^2 - 1$ and the second one among all $(q^2 - 1) - (q - 1) = q^2 - q$ non-proportional ones. Therefore, there are $\frac{(q^3-1)(q^3-q)}{(q^2-1)(q^2-q)} = q^2 + q + 1$ distinct planes in $\mathbb{F}_q^3$. □

The reason why finite projective planes and the Dobble game are isomorphic can be now understood. For $q = 7$, Lemma 4.3 tells that in $\mathbb{P}_7^2$ there are $q^2 + q + 1 = 57$ distinct points and lines (notice that the real Dobble game,

with 55 cards, is actually defective of 2 cards!). By Lemma 4.2 every projective line in $\mathbb{P}_7^2$ has $q+1=8$ points in it, and every point is on $q+1=8$ projective lines. Axioms 1 and 2 for projective planes ensure that, by identifying projective points with symbols and projective lines with cards, the resulting game has the Dobble property. We have in fact that, every card has $q+1=8$ symbols in it, each symbol is on $q+1=8$ cards, and any two cards share one symbol.

5 Generating Dobble games

The general discussion about finite projective planes implies that for any $q=p^m$, with p prime and $m \geq 1$, a Dobble game with q^2+q+1 symbols and cards, and with $q+1$ symbols on each card, can be constructed. However, given q^2+q+1 distinct symbols, we are still left with no machinery to actually *construct* a Dobble game with those symbols. In the following, an explicit procedure to arrange the q^2+q+1 symbols of a general Dobble game of prime order q into its cards is provided. From now on card-line and symbol-point terms are used interchangeably.

Remark 1. *Let $q=p^m$ prime power, with $m>1$. In the following we are heavily using the identification $\mathbb{F}_q = \overline{q} = \{0,\ldots,q-1\}$ which allows us to preserve the field structure for integer operations via the $\varphi(\psi(.))$ bijection.*

We start with q^2+q+1 distinct symbols. Let's pick one card L and name its $q+1$ symbols $L=\{b_0,b_1,b_2,\ldots,b_q\}$. The symbol b_0 is on $q+1$ cards, thus let us list the other q cards L_i^0, for $i=0,1,\ldots,q-1$, that share (only) the symbol b_0 with L:

$$\begin{array}{rccccc} L: & b_0 & b_1 & b_2 & \ldots & b_q \\ L_0^0: & b_0 & a_{0,0} & a_{0,1} & \ldots & a_{0,q-1} \\ L_1^0: & b_0 & a_{1,0} & a_{1,1} & \ldots & a_{1,q-1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ L_{q-2}^0: & b_0 & a_{q-2,0} & a_{q-2,1} & \ldots & a_{q-2,q-1} \\ L_{q-1}^0: & b_0 & a_{q-1,0} & a_{q-1,1} & \ldots & a_{q-1,q-1} \end{array} \tag{1}$$

We named the remaining symbols so they are arranged in a matrix $A=(a_{i,j})$, for $i,j=0,1,\ldots,q-1$. By assumption all $a_{i,j}$ are distinct, thus we have listed in (1) all q^2+q+1 different symbols. For any other symbol b_k, with $k=1,\ldots,q$, let us denote by L_ℓ^k, with $\ell=0,1,\ldots,q-1$, the cards that share the symbol b_k. Each card L_ℓ^k must be intersecting with all cards L_i^0, for all $i=0,1,\ldots,q-1$, in one (and only one) symbol. Therefore, each card L_ℓ^k must contain one symbol from each row i (at unknown ?-th column) of matrix A. Compactly written:

$$\begin{array}{rccccc} L_0^k: & b_k & a_{0,?} & a_{1,?} & \ldots & a_{q-1,?} \\ L_1^k: & b_k & a_{0,?} & a_{1,?} & \ldots & a_{q-1,?} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ L_{q-2}^k: & b_k & a_{0,?} & a_{1,?} & \ldots & a_{q-1,?} \\ L_{q-1}^k: & b_k & a_{0,?} & a_{1,?} & \ldots & a_{q-1,?} \end{array} \tag{2}$$

Let us name A^k the inner matrix of values of the form $a_{i,?}$ for the given k. By uniqueness of symbols, we have seen that each column j of A^k must be

permutation of row j of A, for $j = 0, \ldots, q-1$. This must hold true for all $k = 1, \ldots, q$. There must therefore exist a family of permutations $\sigma[k] : \overline{q} \times \overline{q} \longrightarrow \overline{q}$ parametrized by $k = 1, \ldots, q$, such that each card can be written as

$$L_\ell^k = \{ \quad b_k \quad , \quad a_{0,\sigma[k](\ell,0)} \quad , \quad \cdots \quad , \quad a_{q-1,\sigma[k](\ell,q-1)} \quad \} \tag{3}$$

and the Dobble property is preserved. This means that we need σ such that:

(i) $\sigma[k](., j) : \overline{q} \longrightarrow \overline{q}$ is a permutation (as observed before);

(ii) $\sigma[k](\ell, .) : \overline{q} \longrightarrow \overline{q}$ is a permutation, such that for any two ℓ_1, ℓ_2:

- $\sigma[k](\ell_1, j) \neq \sigma[k](\ell_2, j)$ for all j, whenever $\ell_1 \neq \ell_2$;
- $\sigma[k](\ell_1, j) \neq \sigma[h](\ell_2, j)$ for all j except one, whenever $k \neq h$.

Such family of permutations exists and it is given explicitly in the following Theorem.

Theorem 5.1. *The family of permutations parametrized by $k = 1, \ldots, q$ given by*

$$\begin{aligned} \sigma[k] : \overline{q} \times \overline{q} &\longrightarrow \overline{q} \\ (\ell, j) &\longmapsto \ell + jk \end{aligned} \tag{4}$$

preserves the Dobble property.

Proof. Any two cards from the $k = 0$ group of cards (those listed in (1)) fulfill the Dobble property by assumption. Let us now intersect any two cards from different groups, thus for any $k, h = 1, \ldots, q$ and $\ell_1, \ell_2 = 0, \ldots, q-1$:

- $L_{\ell_1}^0 \cap L_{\ell_2}^k$. The two cards are

  $$\begin{aligned} L_{\ell_1}^0 &= \{ \quad b_0 \quad , \quad a_{\ell_1,0} \quad , \quad \cdots \quad , \quad a_{\ell_1,q-1} \quad \} \\ L_{\ell_2}^k &= \{ \quad b_k \quad , \quad a_{0,\ell_2+0k} \quad , \quad \cdots \quad , \quad a_{q-1,\ell_2+(q-1)k} \quad \} \end{aligned}$$

  The intersecting symbol is exactly the only one with row index ℓ_1 in $L_{\ell_2}^k$, namely $a_{\ell_1,\ell_2+\ell_1 k}$.

- $L_{\ell_1}^k \cap L_{\ell_2}^k$ with $\ell_1 \neq \ell_2$. The two cards are

  $$\begin{aligned} L_{\ell_1}^k &= \{ \quad b_k \quad , \quad a_{0,\ell_1+0k} \quad , \quad \cdots \quad , \quad a_{q-1,\ell_1+(q-1)k} \quad \} \\ L_{\ell_2}^k &= \{ \quad b_k \quad , \quad a_{0,\ell_2+0k} \quad , \quad \cdots \quad , \quad a_{q-1,\ell_2+(q-1)k} \quad \} \end{aligned}$$

  The only intersecting element is b_k. If otherwise, there would be some $j \in \overline{q}$ such that $\ell_1 + jk = \ell_2 + jk$, then by cancellation we would absurdly have $\ell_1 = \ell_2$.

- $L_{\ell_1}^k \cap L_{\ell_2}^h$ with $k \neq h$. The two cards are

  $$\begin{aligned} L_{\ell_1}^k &= \{ \quad b_k \quad , \quad a_{0,\ell_1+0k} \quad , \quad \cdots \quad , \quad a_{q-1,\ell_1+(q-1)k} \quad \} \\ L_{\ell_2}^h &= \{ \quad b_h \quad , \quad a_{0,\ell_2+0h} \quad , \quad \cdots \quad , \quad a_{q-1,\ell_2+(q-1)h} \quad \} \end{aligned}$$

  The intersecting symbol must satisfy $\ell_1 + jk = \ell_2 + jh$ for one and only one $j \in \overline{q}$. Without loss of generality let us assume $k - h > 0$, then the condition becomes $(\ell_1 - \ell_2) + j(k-h) = 0$, which is verified by one and only one $j \in \overline{q}$ because of the field structure of $\overline{q}$, where the multiplicative inverse is unique.

□

6 Numerical implementation

Here is the Python code which implements the procedure outlined in Section 5. Finite field arithmetic is managed by the `galois` package. The finite field `F=galois.GF(Q)` is first instantiated, then the core formula defined in Theorem 5.1 is computed in `F` arithmetic via `F(l)+F(j)*F(k)`.

6.1 Python code

```
import sys
import numpy as np
import galois
np.set_printoptions(threshold=sys.maxsize)

def dobble(p, m):
  Q = p**m          # order of the finite field
  N = Q*(Q+1)+1     # number of points and lines (symbols and cards)
  F = galois.GF(Q)  # initializing the Galois Field of order Q=p^m
  print(f'Dobble of order:   {Q}')
  print(f'Symbols per card:  {Q+1}')
  print(f'Number of cards:   {N}')

  #Symbol-set S of N elements indexed linearly
  dform = '{{:0{}d}}'.format(len(str(N)))
  #S[i] for i=0,...,N-1 is a symbol
  S = np.array([dform.format(s) for s in range(1, N+1)])
  #Print the symbols
  print('Symbols: \n', S)
  #The first Q+1 symbols are the "b_0, b_1, ..., b_Q"
  B = np.arange(0, Q+1)
  #Subsequent symbols are wrapped into matrix A
  A = np.arange(Q+1, N).reshape((Q, Q))
  #List of arrays indexing the symbols set of each card
  index = []
  #Fill matrix A0 by rows
  index.append(B)
  for l in range(0, Q):
    row = [ B[0], *[ A[l, j] for j in range(0, Q) ] ]
    index.append(row)
  #Fill matrices A1, A2, ..., AQ
  for k in range(1, Q+1):
    #Fill matrix by rows
    for l in range(0, Q):
      row = [ B[k], *[ A[j, F(l)+F(j)*F(k-1)] for j in range(0, Q) ] ]
      index.append(row)
  #Convert indexes to symbols
  deck = np.array([S[q] for q in index])
  #Print the Dobble deck
  print('Deck: \n',deck)
```

6.2 Dobble game of order $p = 3$

As an example for a prime order Dobble game let $p = 3$ and $m = 1$.

```
p = 3
m = 1
dobble(p, m)
```

The corresponding Dobble game has 13 symbols and cards, with 4 symbols on each card.

```
Dobble of order:   3
Symbols per card:  4
Number of cards:   13
Symbols:
 ['01' '02' '03' '04' '05' '06' '07' '08' '09' '10' '11' '12' '13']
Deck:
 [['01' '02' '03' '04']
 ['01' '05' '06' '07']
 ['01' '08' '09' '10']
 ['01' '11' '12' '13']
 ['02' '05' '08' '11']
 ['02' '06' '09' '12']
 ['02' '07' '10' '13']
 ['03' '05' '09' '13']
 ['03' '06' '10' '11']
 ['03' '07' '08' '12']
 ['04' '05' '10' '12']
 ['04' '06' '08' '13']
 ['04' '07' '09' '11']]
```

6.3 Dobble game of order $q = 2^2$

As an example for a prime power order Dobble game let $p = 2$ and $m = 2$.

```
p = 2
m = 2
dobble(p, m)
```

The corresponding Dobble game has 21 symbols and cards, with 5 symbols on each card.

```
Dobble of order:   4
Symbols per card:  5
Number of cards:   21
Symbols:
 ['01' '02' '03' '04' '05' '06' '07' '08' '09' '10' '11' '12' '13' '14'
 '15' '16' '17' '18' '19' '20' '21']
```

```
Deck:
 [['01' '02' '03' '04' '05']
 ['01' '06' '07' '08' '09']
 ['01' '10' '11' '12' '13']
 ['01' '14' '15' '16' '17']
 ['01' '18' '19' '20' '21']
 ['02' '06' '10' '14' '18']
 ['02' '07' '11' '15' '19']
 ['02' '08' '12' '16' '20']
 ['02' '09' '13' '17' '21']
 ['03' '06' '11' '16' '21']
 ['03' '07' '10' '17' '20']
 ['03' '08' '13' '14' '19']
 ['03' '09' '12' '15' '18']
 ['04' '06' '12' '17' '19']
 ['04' '07' '13' '16' '18']
 ['04' '08' '10' '15' '21']
 ['04' '09' '11' '14' '20']
 ['05' '06' '13' '15' '20']
 ['05' '07' '12' '14' '21']
 ['05' '08' '11' '17' '18']
 ['05' '09' '10' '16' '19']]
```